Group and solo flights

Does flying in a group change how a paraglider moves?

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23 October 2026 • analysis: `xc-lab/analysis/groups/group-identification` • grouping run `d120_eh500_ez300_th45_g120_q0.1`

Abstract

Do paragliders move differently when they fly in a group? Group time differs from solo flight in many ways, but so do the pilots who fly in groups, the places and hours where groups form, and the moments pilots choose to fly together. We measure the motion by the local exponent $H_{\mathrm{loc}}(\tau)$ of the time-averaged mean squared displacement and by statistics of the climbs and glides, and take each of the other causes away with one test: contrasts within strata of site, day and hour and within the same flight; solo flights reweighted to the zones and hours of the groups; the same flight before, during and after a group. Over a quarter of an hour a pilot in a group progresses straighter: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s}) = +0.037$ within the same flight and $+0.043$ at the same site, day and hour, of which who flies in groups accounts for $+0.003$. The zone and the hour remove at most 19% of the gap, and on the same flight the effect starts and stops with the group. Within one glide or climb nothing changes. In a group pilots also climb more, end more searches in a climb, and glide faster and flatter. Most of this is there with two pilots, and the pilot ahead finds no more thermals than the one behind. The tests fix the pilot, the place, the hour and the moment, but not what brings pilots together: the effect goes with flying together, and whether the group causes it stays open.

1 Introduction

The thesis studies how paraglider pilots move in cross-country flight: how long they climb, how far they glide, how their displacement grows with time. Flying in a group could change all of this. In soaring birds, social information on the position of thermals changes the gliding strategy, and in a migrating flock a few individuals find the thermals while the others follow them. The note *Group flights in free flight* (`ideas/group-identification`) labels every instant of every flight of the FFVL archive as solo or in a group. This report asks whether, on the same archive, pilots move differently when they fly in a group.

A raw comparison cannot answer, because group time differs from solo flight in four other ways.

(i) **The pilots.** Those who fly in groups may be more experienced, or fly faster wings.
(ii) **The place.** Groups form on some sites and routes, where any pilot may progress straighter.
(iii) **The day and the hour.** Group time gathers around midday and on some days more than others, and the air of that day and hour acts on every pilot.
(iv) **The moment.** Pilots may join others when the progress is straight anyway, along a convergence line or a ridge, and the straight progress would then come before the group.

Each of these produces a difference between group and solo time with no effect of the group. The report removes them one at a time. Section 2 fixes the populations and the measures, and Section 3 shows the raw difference. Contrasts within strata of site, day and hour, and within the same flight, separate the pilots and the day from the group (Section 4); reweighting the solo flights to the zones and hours where groups fly tests the place (Section 5); the same flight before, during and after a group tests the moment (Section 6). What survives the three tests we call the group effect. Section 7 looks inside the group, at its size and at who leads, and Section 8 says what can and cannot be concluded.

2 Data and measures

Which flights, which populations, and which measures of the motion?

Flights and groups. 156 391 paraglider and 6 094 hang-glider flights on 6 209 days, the cleaned tracks of the old repository (soaring, tag `legacy-final`). A flight is analysed after its *departure*, the first fix 3 km from the takeoff; before it the pilot counts as neither solo nor in a group. The measures of the motion use loggers

of at most 10 s; the phases (climb, search, glide) are Vilpellet's runs and exist on the flights logged every second, 70% of the paragliders. The groups are those of the note, with $\delta = 2$ min, $\varepsilon_h = 500$ m, $\varepsilon_z = 300$ m, $\theta_{\max} = 45°$, $g = 2$ min and $q = 0.1$, hence contacts of at least $\tau^* = 38.1$ min (its sections 3–6). A contact means shared air within sight; it does not prove interaction.

Populations. Three populations of stretches of flight.

- S, *solo flights*: flights never in a group.
- G, *group time*: the stretches of a flight inside a configuration, a fixed set of flights that stays one group over a maximal interval. G_n is the group time in configurations of n pilots; "n pilots" always means the size of the configuration at that instant.
- A, *the same flights alone*: the stretches of the group flights outside any configuration. It has the pilots, gliders, loggers and days of G, and serves as its control.

The motion over every lag. The horizontal position $\mathbf{r}(t)$ of a flight is read on a 10-s grid of its clock. For a flight f and a population, the time-averaged mean squared displacement pools the origins of all its stretches in that population,

$$\overline{\delta_f^2}(\tau) = \frac{1}{N_f(\tau)} \sum_t \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2, \tag{1}$$

where t runs over the $N_f(\tau)$ origins with t and $t+\tau$ in the same stretch, and the ensemble $\mathrm{MSD}(\tau)$ is the mean of $\overline{\delta_f^2}(\tau)$ over the flights, one weight per flight. A stretch enters only if it lasts at least 1250 s, so that every population keeps the same flights at all 21 lags from 10 to 1000 s. With $\mathrm{MSD}(\tau) \propto \tau^{2H}$, H is half the slope of $\log \mathrm{MSD}$ against $\log \tau$ over all lags: $H = 1$ for a straight line, $H = 1/2$ for a random walk. $H_{\mathrm{loc}}(\tau)$ is the same slope over the lags within ± 0.25 decades of τ (Appendix A). We read it at two scales: $\tau = 100$ s, within one climb or one glide, and $\tau = 1000$ s, over several climb–glide cycles.

Climbs and glides. Each phase run goes whole to the population of the stretch that holds its middle, and each statistic is a ratio of sums over the runs of a population, so long phases weigh more. Of the twelve statistics measured, five are not explained by the others and are the ones drawn (Appendix C): the share of time in climb, the share of searches that end in a climb, the climb rate (total height gained over total climb time), the glide speed and the glide ratio, replaced in stratified contrasts by its inverse, the height lost per kilometre of glide (Appendix A). The tables also give the cross-country speed, the straight-line distance covered in 30 min averaged over every origin, which the five explain.

Strata. A *zone* is a cell of 25 km containing the middle of a piece of track; a *site-day* holds the flights of one day whose takeoffs lie within 3 km; the *hour* is the local hour of the middle of a piece or a run.

Uncertainty. Bands are 5–95% of bootstrap draws that resample whole dependency blocks, the flights tied by sharing episodes, and recompute the statistic: the blocks of the note for pooled statistics (1000 draws), blocks that also join the flights of a site-day for stratified ones (200 draws, the same for every statistic). They measure the sampling of blocks, not the choice of q or of the tolerances. Two bands that do not overlap are not a test; differences are computed draw by draw.

What the grouping gives. Of the 154 598 paraglider flights with a logger of at most 10 s, 33 501 (21.7%) fly in a group at some point; of the 6 039 hang-glider flights, 418 (6.9%). Group time is 12.8% of all paraglider flight time after the departure, and on average 45% of the time of a group flight. The 36 575 pair intervals last 40 to 96 min (10th to 90th percentile; median 52 min) and form 40 051 configurations in 17 169 episodes. Half of the episodes have two flights, 1% have more than 9, the largest has 53. By the size of the configuration, the paraglider group time is: 2 pilots 64%, 3 pilots 19%, 4 pilots 8%, 5-6 pilots 6%, 7+ pilots 3% (the largest configuration has 27 pilots). In contact, one pilot is behind the other by a median of 20 s (75% within 45 s).

What we keep. Three populations, S, G and A; two scales of the motion, $H_{\mathrm{loc}}(100\ \mathrm{s})$ and $H_{\mathrm{loc}}(1000\ \mathrm{s})$; the statistics of climbs and glides; bands from blocks of flights that are not independent.

3 The raw difference

How do group time and solo flights differ before any control?

	solo flights S	group time G	same flights alone A
flights (TA-MSD)	119 129	33 435	29 611
H (10–1000 s)	0.899 [0.898, 0.899]	0.916 [0.915, 0.917]	0.914 [0.913, 0.915]
$H_{\mathrm{loc}}(100\ \mathrm{s})$	0.919 [0.919, 0.919]	0.927 [0.927, 0.928]	0.932 [0.931, 0.932]
$H_{\mathrm{loc}}(1000\ \mathrm{s})$	0.837 [0.837, 0.838]	0.893 [0.891, 0.894]	0.860 [0.859, 0.862]
time in climb (%)	30.8 [30.7, 30.9]	33.0 [32.8, 33.2]	29.1 [28.9, 29.2]
searches ending in a climb (%)	43.3 [43.2, 43.4]	49.5 [49.2, 49.7]	45.9 [45.7, 46.1]
climb rate (m/s)	1.22 [1.22, 1.22]	1.26 [1.25, 1.27]	1.33 [1.32, 1.34]
glide speed (km/h)	30.1 [30.1, 30.2]	33.3 [33.1, 33.5]	32.8 [32.6, 32.9]
glide ratio	10.4 [10.4, 10.4]	12.9 [12.8, 13.0]	11.3 [11.2, 11.4]
cross-country speed (km/h)	17.2 [17.2, 17.3]	20.0 [19.8, 20.3]	20.0 [19.8, 20.1]

Table 1: Pooled statistics of S, G and A (paragliders), with 5–95% bands. Exponents on loggers of at most 10 s, phases on 1 Hz flights.

Group time progresses straighter over long lags than solo flights, $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s}) = +0.055$; it also spends more time climbing, ends more searches in a climb and glides faster (Table 1; the curves over every lag are in Appendix D). The same flights alone sit between the two at long lags and have the highest climb rate. Part of the raw difference may therefore belong to who flies in groups, and to where and when they fly.

What we keep. A difference to explain, largest at long lags. The next three sections ask how much of it is the group.

4 Group or pilots? Contrasts within strata

How much of the difference belongs to who flies in groups, and how much to the group, with the day, the site and the hour fixed?

Contrast. For a statistic X and two populations P and Q, each stratum s that holds both gives $D_s = X_{P,s} - X_{Q,s}$, computed as on the whole archive (ensemble MSD, ratios of sums). The contrast is their weighted mean,

$$\Delta_{P-Q} = \frac{\sum_s w_s D_s}{\sum_s w_s}, \qquad w_s = \frac{n_{P,s}\, n_{Q,s}}{n_{P,s} + n_{Q,s}}, \tag{2}$$

with $n_{P,s}$ and $n_{Q,s}$ the flights of each population in s. The weight is half the harmonic mean of the two counts: a stratum counts as much as its smaller side allows. For the exponents, D_s is the difference of log MSD at every lag, and ΔH and ΔH_{loc} are the slopes of its weighted mean.

Five contrasts. Each fixes something more (Table 2):

- G − S pooled: the raw difference;
- G − S within site-day and hour: the same day, site and hour;
- A − S within site-day and hour: *selection*, how the flights that will fly in a group differ when they fly alone;
- G − A within the flight, the stratum being one flight: pilot, glider, logger, day and site fixed;
- G − A within site-day and hour: the hour fixed as well.

The last two estimate the group effect and fail in opposite ways. Within the flight, group time falls around midday and the time alone of the same flight earlier and later (section 7 of the note), so the hour differs. Within site-day and hour, the hour is the same but the pilots differ. Where the two agree, the effect belongs to the group.

Results. The straighter progress over 1000 s survives every control: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.055$ pooled, $+0.037$ within the flight and $+0.043$ at the same site-day and hour, while selection gives $+0.003$. Within one phase nothing remains: at 100 s, -0.009 within the flight and $+0.002$ at the same site-day and hour; the pooled $+0.008$ was selection and composition. In a group pilots climb more ($+5.71$ and $+3.28$ points of time in climb), end more searches in a climb ($+8.75$ and $+2.59$ points) and climb faster ($+0.024$ and $+0.029$ m/s). Pooled, the climb rate of G is below that of A: the pooled ratios weight flights by their time in each population, and the strata undo that. Glides are faster, $+1.36$ and $+0.87$ km/h, and flatter, -21.1 and -9.5 m/km of height lost per km; selection adds $+1.15$ km/h of glide speed, so group pilots glide faster even alone. The cross-country speed rises by $+0.70$ and $+1.29$ km/h.

What we keep. With the pilot, the day, the site and the hour fixed, a group progresses straighter over a quarter of an hour, climbs more and glides faster and flatter. Who flies in groups explains little of it.

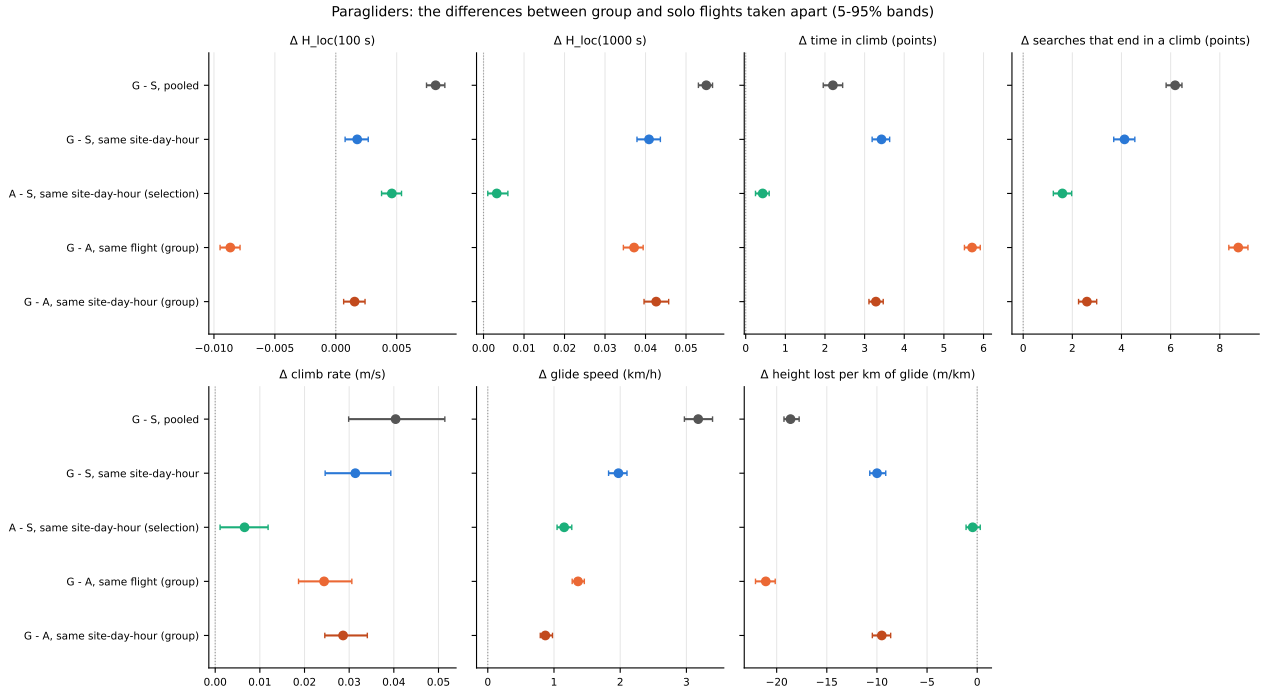


Figure 1: Each statistic (panel) for the five contrasts (rows), with 5–95% bands. Shares of time in points; lower height lost per km is a better glide.

Δ	G − S pooled	G − S same s-d-h	A − S same s-d-h	G − A same flight	G − A same s-d-h
$H_{\text{loc}}(100\text{ s})$	+0.008 [+0.007, +0.009]	+0.002 [+0.001, +0.003]	+0.005 [+0.004, +0.005]	−0.009 [−0.009, −0.008]	+0.002 [+0.001, +0.002]
$H_{\text{loc}}(1000\text{ s})$	+0.055 [+0.053, +0.057]	+0.041 [+0.038, +0.044]	+0.003 [+0.001, +0.006]	+0.037 [+0.035, +0.039]	+0.043 [+0.040, +0.046]
time in climb (%)	+2.20 [+1.96, +2.45]	+3.43 [+3.19, +3.63]	+0.43 [+0.25, +0.59]	+5.71 [+5.52, +5.92]	+3.28 [+3.11, +3.47]
searches to climb (%)	+6.18 [+5.82, +6.46]	+4.12 [+3.68, +4.54]	+1.60 [+1.23, +1.97]	+8.75 [+8.37, +9.15]	+2.59 [+2.25, +2.99]
climb rate (m/s)	+0.040 [+0.030, +0.051]	+0.031 [+0.025, +0.039]	+0.007 [+0.001, +0.012]	+0.024 [+0.019, +0.031]	+0.029 [+0.025, +0.034]
glide speed (km/h)	+3.18 [+2.97, +3.39]	+1.97 [+1.82, +2.10]	+1.15 [+1.05, +1.27]	+1.36 [+1.27, +1.46]	+0.87 [+0.79, +0.98]
height lost per km (m/km)	−18.6 [−19.3, −17.8]	−10.0 [−10.7, −9.1]	−0.5 [−1.1, +0.3]	−21.1 [−22.1, −20.1]	−9.5 [−10.5, −8.6]
cross-country speed (km/h)	+2.81 [+2.59, +3.01]	+2.25 [+2.09, +2.39]	+1.31 [+1.21, +1.43]	+0.70 [+0.60, +0.78]	+1.29 [+1.16, +1.41]

Table 2: The contrasts of Figure 1: difference and 5–95% band (s-d-h: same site-day and hour).

5 Group or place? Solo flights where groups fly

Do groups fly where, and when, any pilot would progress straighter? If so, solo flights compared on the same ground and at the same hours would match group time.

Reweighting. Each piece of track gets a stratum from its middle, from coarse to fine: zone, zone×month, zone×hour, site-day×hour. The solo flights are then given the mix of the group over the strata (direct standardisation):

$$\text{MSD}_{\text{S}|\text{G}}(\tau) = \frac{\sum_s n_{\text{G},s}\,\text{MSD}_{\text{S},s}(\tau)}{\sum_s n_{\text{G},s}}, \tag{3}$$

with $n_{\text{G},s}$ the units of G in stratum s (a flight counts once per stratum it crosses) and $\text{MSD}_{\text{S},s}$ the ensemble MSD of the solo units of s. In the statistic, S then flies where and when G flies. G is taken on the same strata, so the two curves share their support; the same is done for A. "As is" is the comparison on the same strata without the reweighting.

Results. The gap opens above $\tau \approx 150$ s (Figure 3): ΔH_{loc} between G and S is +0.005 to +0.011 from 20 to 140 s and +0.055 at 1000 s, above one climb–glide cycle. The group therefore acts on how the cycles are

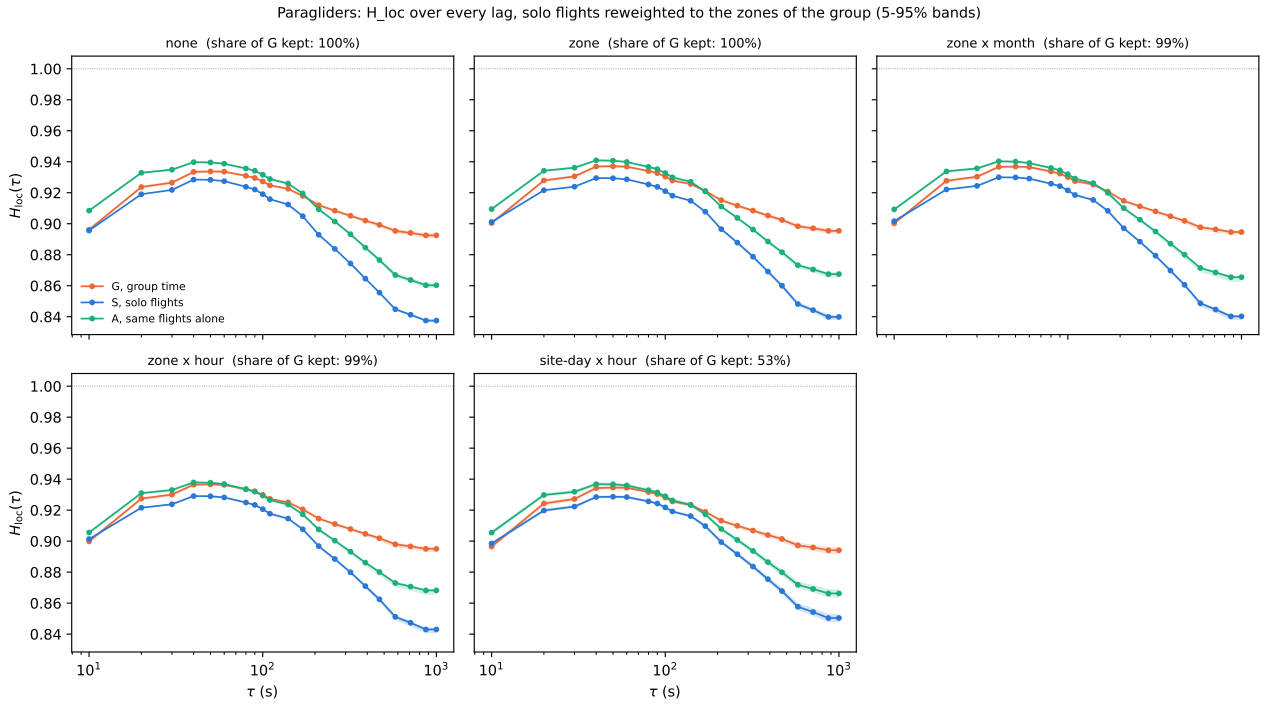


Figure 2: $H_{\mathrm{loc}}(\tau)$ of G (orange), of S reweighted to the strata of G (blue) and of A reweighted to them (green), 5–95% bands. First panel: all flights pooled. "Share of G kept": flights of G with a solo counterpart in their stratum.

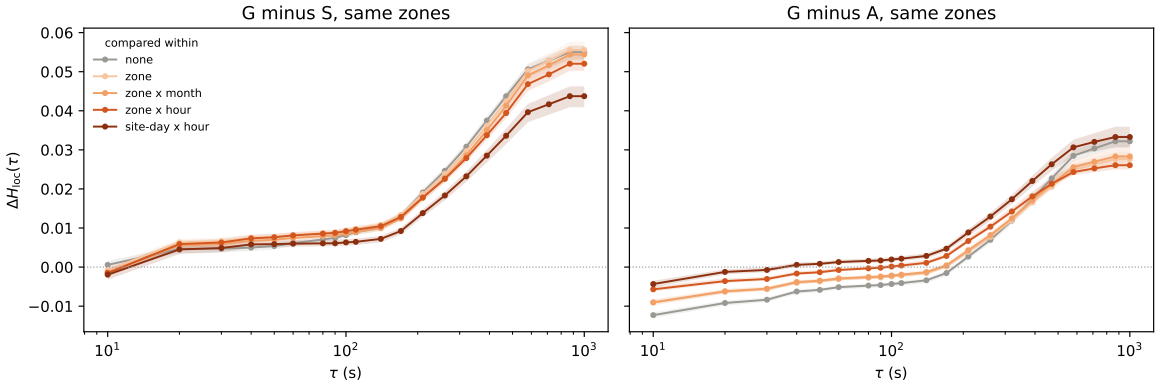


Figure 3: $\Delta H_{\mathrm{loc}}(\tau)$ over every lag: G minus S (left) and G minus A (right), the other population reweighted to the strata of G; one line per stratification, finer strata in darker orange.

		fitted H (10–1000 s)			ΔH			$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$		
compared within	G kept (%)	G	S | G	A | G	G − S as is	G − S reweighted	G − A reweighted	G − S as is	G − S reweighted	G − A reweighted
none	100	0.916	0.899	0.914	+0.017	+0.017 [+0.016, +0.018]	+0.002 [+0.001, +0.003]	+0.055	+0.055 [+0.053, +0.057]	+0.032 [+0.030, +0.034]
zone	100	0.919	0.902	0.916	+0.021	+0.018 [+0.017, +0.019]	+0.003 [+0.003, +0.004]	+0.059	+0.056 [+0.054, +0.058]	+0.028 [+0.026, +0.029]
zone × month	99	0.919	0.902	0.915	+0.022	+0.017 [+0.016, +0.018]	+0.003 [+0.003, +0.004]	+0.064	+0.054 [+0.053, +0.057]	+0.028 [+0.027, +0.031]
zone × hour	99	0.919	0.902	0.913	+0.022	+0.017 [+0.016, +0.018]	+0.005 [+0.005, +0.006]	+0.064	+0.052 [+0.050, +0.054]	+0.026 [+0.025, +0.028]
site-day × hour	53	0.917	0.904	0.913	+0.015	+0.013 [+0.012, +0.014]	+0.008 [+0.007, +0.009]	+0.051	+0.044 [+0.041, +0.046]	+0.033 [+0.031, +0.036]

Table 3: Fitted H and differences by stratification. "Reweighted": the other population reweighted to the strata of G (value, 5–95% band); "as is": the same strata without reweighting. $H_{\mathrm{S|G}}$: H of S reweighted to G.

chained, while a single glide or climb is unchanged. Reweighting to the zones of G lowers $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ by 6% ($+0.059 \to +0.056$) and the fitted ΔH by 15%; adding the month, by 15% and 23%; adding the hour, by 19% and 24%. Solo pilots do fly where groups fly: 100% of G has a solo counterpart in its cell. At the same site-day and hour (53% of G kept), $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s}) = +0.044$ [$+0.041$, $+0.046$], 14% below its value as is, and the fitted $\Delta H = +0.013$ [$+0.012$, $+0.014$]: reweighting raises the fitted H of S by a few thousandths ($0.899 \to 0.904$) and never brings it to that of G (0.917). Against the same flights alone, $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.032$ pooled and $+0.028$, $+0.026$ and $+0.033$ in zone, zone×hour and site-day×hour strata. Below 150 s the sign of G − A depends on the hour, which is why the two estimates of Section 4 are read together.

Limits. Cells are 25 km, placed by the middle of a piece; the terrain inside a cell is not controlled, and a group may choose better air within it, which is then part of what we call the group effect. Paragliders only.

What we keep. The place and the hour where groups fly explain at most a fifth of the gap; the rest stays at the same site, day and hour.

6 Group or moment? The same flight before, during and after

Do pilots fly together in the moments when the progress is straight anyway? If so, the same flight would already be straight just before the group forms, and still just after it breaks.

Windows. Every stretch of group time lasting at least $W = 50$ min is an *event*, with up to four windows of length W on the clock of the same flight: *before*, the last W of the solo stretch that precedes it; *start* and *end*, the first and last W of the group; *after*, the first W of the solo stretch that follows it. Before and after exist only when a solo stretch at least W long touches the group. For two windows the ensemble MSD is taken on the events that have both, and their difference is read on H_{loc} and on the phase statistics. The two readings predict different things:

- if the straighter progress belongs to the group, H_{loc} rises from before to start, falls from end to after, and before and after are alike;
- if it belongs to the moment, before is already high, and the changes are those of any two consecutive windows.

Controls. Two consecutive windows with no change of state, drawn in the solo time of the group flights (A) and in solo flights (S), measure how much a flight drifts from one window to the next.

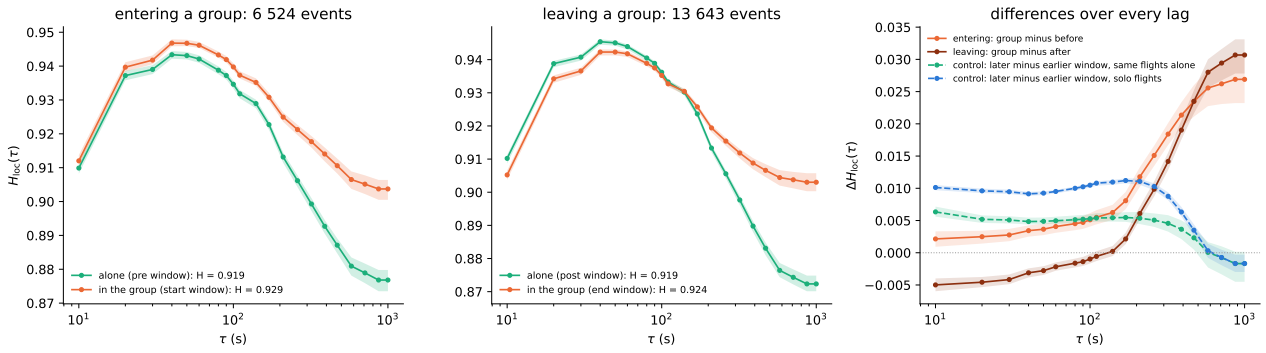


Figure 4: $H_{\mathrm{loc}}(\tau)$ of the same flights alone (green) and in the group (orange), entering (left) and leaving (middle), 5–95% bands. Right: the differences, with the two controls (later minus earlier window; dashed).

Results. The first prediction holds. $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.027$ [$+0.023$, $+0.031$] entering and $+0.031$ [$+0.028$, $+0.033$] leaving, against -0.002 and -0.002 in the controls, and after minus before is -0.003 [-0.007, $+0.001$]. On the 3 334 events with all four windows, $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.879 before, 0.910 and 0.911 in the group and 0.876 after (Figure 5); the step matches the within-flight estimate of Section 4 ($+0.037$). At 100 s the changes, $+0.005$ entering and -0.001 leaving, lie inside the drift of the controls ($+0.005$, $+0.010$). The fitted H ($+0.010$ and $+0.005$ against $+0.005$ and $+0.009$) does not separate from the drift either, because its slope is set mostly by the lags below 150 s. Glides are faster and flatter in the group in both directions: glide speed $+0.92$ km/h entering and $+0.52$ leaving, height lost per km -2.3 and -10.9 m/km, against $+0.04$ and $+0.07$ km/h, $+3.3$

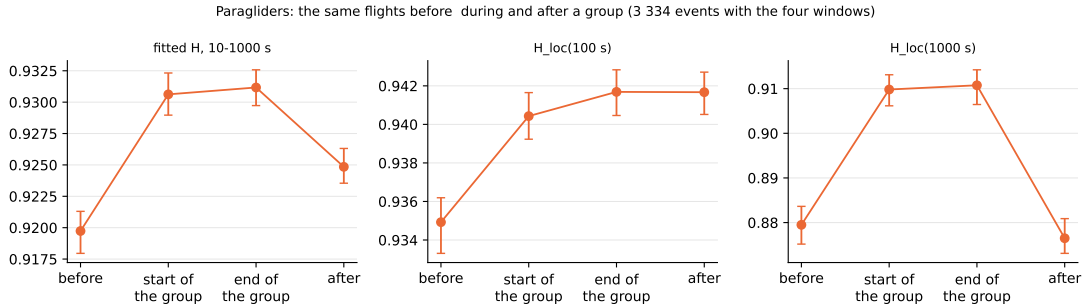


Figure 5: The four windows on the events that have all of them. Start and end overlap when the group lasts less than $2W$.

comparison	events	ΔH	$\Delta H_{\text{loc}}(100)$	$\Delta H_{\text{loc}}(1000)$
entering: start − before	6 524	+0.010 [+0.009, +0.011]	+0.005 [+0.004, +0.006]	+0.027 [+0.023, +0.031]
leaving: end − after	13 643	+0.005 [+0.004, +0.006]	−0.001 [−0.002, −0.000]	+0.031 [+0.028, +0.033]
after − before	6 320	+0.004 [+0.003, +0.005]	+0.006 [+0.005, +0.007]	−0.003 [−0.007, +0.001]
control A: later − earlier	16 998	+0.005 [+0.004, +0.005]	+0.005 [+0.005, +0.006]	−0.002 [−0.005, +0.000]
control S: later − earlier	69 648	+0.009 [+0.008, +0.009]	+0.010 [+0.010, +0.011]	−0.002 [−0.003, −0.000]

Table 4: Differences of the fitted H and of H_{loc} at 100 and 1000 s (value, 5–95% band) between windows, on the events with both windows. Controls: two consecutive windows with no change of state, in the solo time of the group flights (A) and in solo flights (S).

and +4.9 m/km in the controls. Time in climb, successful searches and climb rate move by the size of the drift.

Limits. The windows fix the pilot, the glider, the day and the slow drift of the flight, not what brings pilots together: a convergence line, a ridge or a common route may last as long as the group and straighten the progress by itself, and two pilots must move alike to stay within 500 m for W. The events are long groups with solo flight beside them (6 524 entries, 13 643 exits). With windows of 21 min the same test is noisier (Appendix G).

What we keep. On the same flight the straighter progress over a quarter of an hour starts with the group and stops with it, and the glides change with it; the moment of joining does not explain them.

7 Inside the group: size and leader

Does the effect grow with the number of pilots, and does the pilot ahead fly differently from the one behind?

7.1 Size of the configuration

The group time is cut where the configuration changes, so every stretch has a fixed number of pilots n (2, 3, 4, 5–6, 7+), and a flight can contribute to several sizes. Two contrasts of Eq. (2) follow the size: $G_n - A$ within

comparison	events	time in climb (points)	searches to climb (points)	climb rate (m/s)	glide speed (km/h)	height lost per km (m/km)
entering: start − before	4 786	+0.7 [+0.3, +1.1]	+0.0 [−1.1, +1.0]	+0.046 [+0.034, +0.062]	+0.92 [+0.69, +1.15]	−2.3 [−3.5, −1.3]
leaving: end − after	10 097	+1.5 [+1.2, +1.9]	+1.2 [+0.4, +1.9]	−0.007 [−0.017, +0.003]	+0.52 [+0.35, +0.64]	−10.9 [−11.8, −10.0]
after − before	4 718	−1.4 [−1.8, −1.0]	−2.8 [−4.0, −1.6]	+0.043 [+0.024, +0.059]	+0.33 [+0.12, +0.55]	+9.1 [+7.8, +10.4]
control A: later − earlier	12 568	−1.1 [−1.3, −0.9]	−1.4 [−2.0, −0.8]	+0.018 [+0.009, +0.027]	+0.07 [−0.05, +0.20]	+3.3 [+2.5, +4.2]
control S: later − earlier	49 092	−1.6 [−1.7, −1.5]	−2.3 [−2.7, −2.0]	+0.011 [+0.008, +0.015]	+0.04 [−0.03, +0.10]	+4.9 [+4.5, +5.3]

Table 5: The phase statistics of the same comparisons (differences with 5–95% band); runs and glides are placed in the window that holds their middle.

the flight, and $G_n - G_2$ at the same site-day and hour, which asks whether a group does more than a pair.

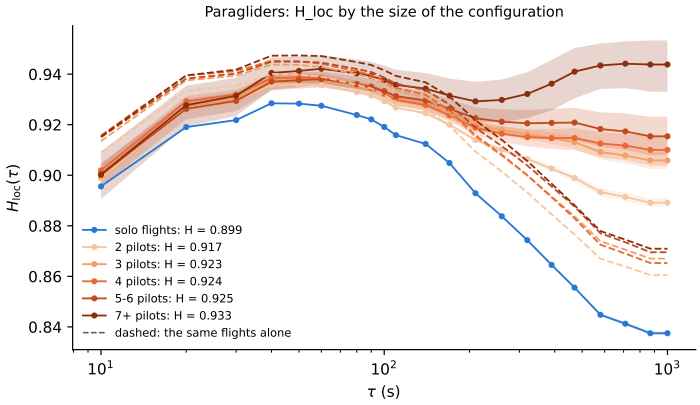


Figure 6: $H_{loc}(\tau)$ over every lag by the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

		H		$H_{loc}(1000\ s)$						
pilots	flights	G_n	alone	G_n	alone	time in climb (%)	searches to climb (%)	climb rate (m/s)	glide speed (km/h)	glide ratio
solo	119 129	0.899		0.837		30.8	43.3	1.22	30.1	10.4
2	26 411	0.917	0.914	0.889	0.861	32.7	49.0	1.27	33.2	12.5
3	9 480	0.923	0.919	0.906	0.867	33.6	49.9	1.26	33.5	13.1
4	4 073	0.924	0.920	0.910	0.865	33.6	51.0	1.25	33.5	13.7
5–6	2 686	0.925	0.922	0.915	0.870	33.1	50.1	1.23	33.1	14.1
7+	1 142	0.933	0.924	0.944	0.871	34.6	51.7	1.20	34.6	15.9

Table 6: Paragliders, pooled by the size of the configuration ("alone": the same flights alone).

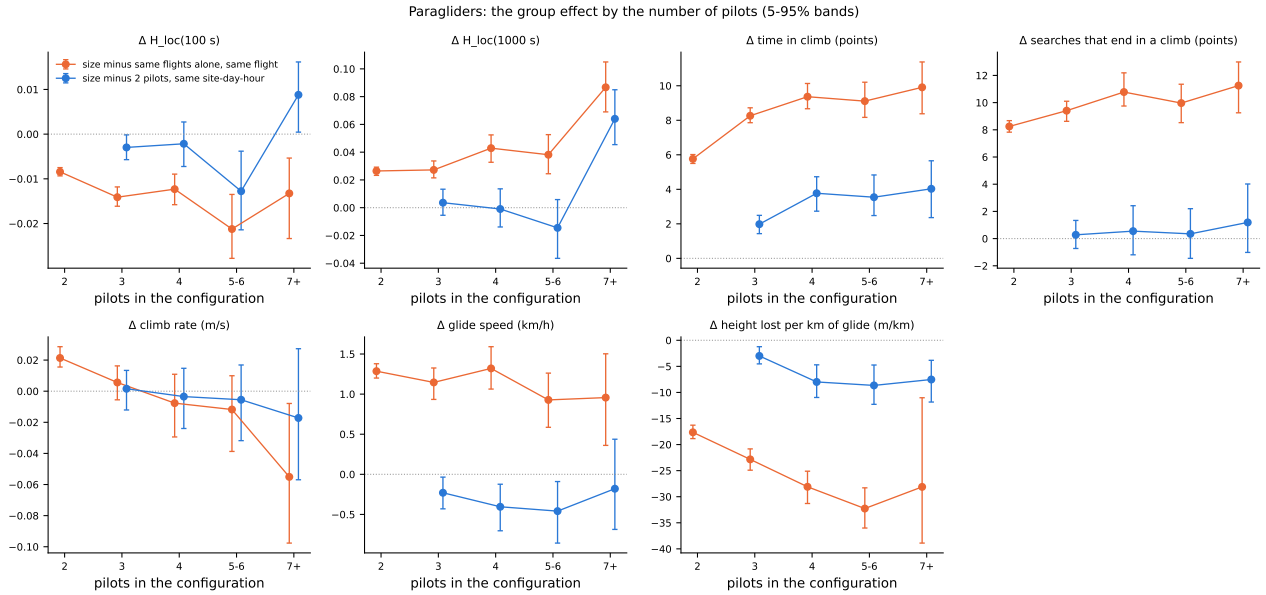


Figure 7: Orange: $G_n - A$ within the flight. Blue: $G_n - G_2$ at the same site-day and hour.

Pooled, H and H_{loc} rise with n while the same flights alone stay flat (Figure 6, Table 6), and within the flight the effect grows with n, from $+0.026$ with 2 pilots to $+0.087$ with 7+. At the same site-day and hour, however, configurations of 3 to 6 pilots do not differ from pairs: $\Delta H_{loc}(1000\ s)$ lies between -0.015 and $+0.004$, with bands across zero (Figure 7). The rise of the pooled curves belongs to the days and sites where large groups form. Only 7+ stands apart ($+0.064$ [$+0.045$, $+0.085$], cross-country speed $+7.0$ km/h), a rare configuration, 3% of the group time. Time in climb and the flatter glide do grow with n at the same place and hour, from $+2.0$ to $+4.0$ points and from -3 to -9 m/km.

7.2 Leader and follower

In configurations of two pilots, the signed median delay ℓ_{med} of the pair interval (Appendix B of the note) says who passes first: the leader, when the other passes at least 15 s later; otherwise the two fly abreast (35% of the pairs; median delay 20 s). Leader and follower are compared within the same configuration, the configuration being the stratum of Eq. (2) (Appendix H). The pilot ahead flies a little faster (glide speed $+0.86$ [$+0.74$, $+0.99$] km/h, climb rate $+0.021$ [$+0.014$, $+0.028$] m/s) but does not find thermals more often (searches ending in a climb -0.01 [-1.03, $+0.83$] points, height lost per km $+1.6$ [-0.9, $+4.5$] m/km): the follower shows no sign of using the leader to find the next thermal. How groups form and break and whether the same pilots lead are in section 7 of the note.

What we keep. Most of the effect is there with two pilots. Beyond a pair, size adds climbing and flatter glides but no straighter progress, except in the rare configurations of 7 or more. Leading changes the speed and leaves the finding of thermals unchanged.

8 What can and cannot be concluded

The raw difference between group time and solo flights is real, and most of it survives the three tests. Fixing the pilot, the day, the site and the hour leaves a straighter progress over a quarter of an hour, more climbing and faster, flatter glides (Section 4); the zones and hours where groups fly explain at most a fifth of the gap (Section 5); on the same flight the effect starts and stops with the group (Section 6). Within one glide or climb nothing changes: the group acts on how the climbs and glides follow each other, at the scale of several cycles. Two pilots carry most of it (Section 7).

Supported

- the progress over a quarter of an hour is straighter in a group, with the pilot, the day, the site and the hour fixed;
- no change within one glide or climb (100 s);
- on the same flight the straighter progress starts and stops with the group, and before and after are alike;
- more time climbing, more searches that end in a climb, a higher climb rate, faster and flatter glides once stratified (the pooled climb rate pointed the other way);
- selection of pilots explains little of the motion;
- the pilot ahead flies a little faster but finds no more thermals than the follower.

Not supported

- that the group *causes* the effect: the tests fix pilot, day, site, hour, zone and the drift of the flight, but not what brings pilots together (a line, a ridge, a route) nor the coordination needed to stay in contact;
- that larger groups straighten the progress, up to six pilots;
- that a contact means interaction: it means shared air within sight;
- the glide statistics are partly fixed by definition, since headings must be aligned in glide for a contact;
- the leader is assigned from ℓ_{med}, which is uncertain in thermals (note, Appendix B).

Open. The phases exist on 1 Hz flights only: the same analyses on the tracks of xc-dynamics, with phases on all loggers, and on hang gliders with enough group flights (Appendix D), remain to be done.

A How H and H_{loc} are computed

$\mathrm{MSD}(\tau) \propto \tau^{2H}$, so $\log_{10}\mathrm{MSD} = 2H\log_{10}\tau + c$. H is half the slope of the least-squares line over all 21 lags (equal weight per lag). $H_{\mathrm{loc}}(\tau_c)$ is half the slope of the line fitted on the lags with $|\log_{10}\tau - \log_{10}\tau_c| \leq 0.25$ (a factor 1.78 on each side), equal weight; where fewer than 3 lags lie in the window, the 3 nearest are used. It is a windowed regression, not a pointwise derivative. ΔH and ΔH_{loc} of two curves are the differences of the fitted values; for a stratified average they are the same regression on the averaged $\Delta\log\mathrm{MSD}$ (linear: slope of a difference = difference of slopes). Bands: the same computation on each bootstrap draw, percentiles lag by lag.

τ (s)	lags used	τ (s)	lags used	τ (s)	lags used
10	10–30 (3 nearest)	90	60–140	320	210–470
20	20–40 (3 nearest)	100	60–170	390	260–580
30	20–50	110	80–170	470	320–710
40	30–60	140	80–210	580	390–1000
50	30–80	170	100–260	710	470–1000
60	40–100	210	140–320	870	580–1000
80	50–140	260	170–390	1000	580–1000

Table 7: Lags of the window of each H_{loc} value. At 10 and 20 s the window is not centred; at 870 and 1000 s it is cut at 1000 s and the two values use the same lags, so the end of the curve flattens by construction. The window resolves nothing narrower than about half a decade.

Phase MSD figures (not in this report) use a different local exponent: the finite-difference derivative of $\log\mathrm{MSD}$ against $\log\Delta$ on 100 lags, without the factor $1/2$ (about 2 for ballistic motion).

Ratios and the glide ratio. Ratios are sums of numerators over sums of denominators over the runs of a population (a run goes to the stretch that holds its middle). In a small stratum a glide in rising air has almost no height lost, the glide ratio diverges and its average over strata is unstable; stratified contrasts use the *height lost per km of glide*, the inverse. Pooled tables keep the glide ratio.

B Checks of the viewer and of the analyses

The checks of the grouping itself, its code against the note and the properties of its tables, are in section 8 of the note.

- **Viewer.** The *Group flights* tab reads `groups/` of its data folder: its seven tables are **byte-identical** to those of the chosen run; the tracks it draws (`SSD_DATA`) have the same fixes, times, positions and trimmed starts as those the grouping read (`SSD_DISANTE`).
- **Analyses.** Tests on synthetic flights for every step, and on every number of the README of the analyses; the decomposition is checked on a synthetic case where the day raises a statistic by 10 and the group by 2: pooled 10, stratified 2.

C Which statistics are drawn

Unit: a population on a site-day with at least 25 flights (119 units). A statistic is redundant when the others explain most of it across units (R^2 of its linear regression on them); backward elimination drops the most explained until every $R^2 \leq 0.5$. The four shares of time sum to 100%, so one of them goes first.

statistic	status	R^2 on the five drawn
time in climb (%)	drawn	0.23
searches that end in a climb (%)	drawn	0.41
climb rate (m/s)	drawn	0.20
glide speed (km/h)	drawn	0.26
glide ratio	drawn	0.38
time in transition (%)	dropped, number 1, R^2 1.00 when dropped	0.83
height gained per climb (m)	dropped, number 2, R^2 0.98 when dropped	0.91
climbs that start after a search (%)	dropped, number 3, R^2 0.93 when dropped	0.77
cross-country speed over 30 min (km/h)	dropped, number 4, R^2 0.93 when dropped	0.90
time in a search in vain (%)	dropped, number 5, R^2 0.88 when dropped	0.71
climbs per hour	dropped, number 6, R^2 0.72 when dropped	0.71
time in a search that ends in a climb (%)	dropped, number 7, R^2 0.65 when dropped	0.65

Table 8: Five statistics remain; the others are explained by them.

D Pooled curves, hang gliders and sizes

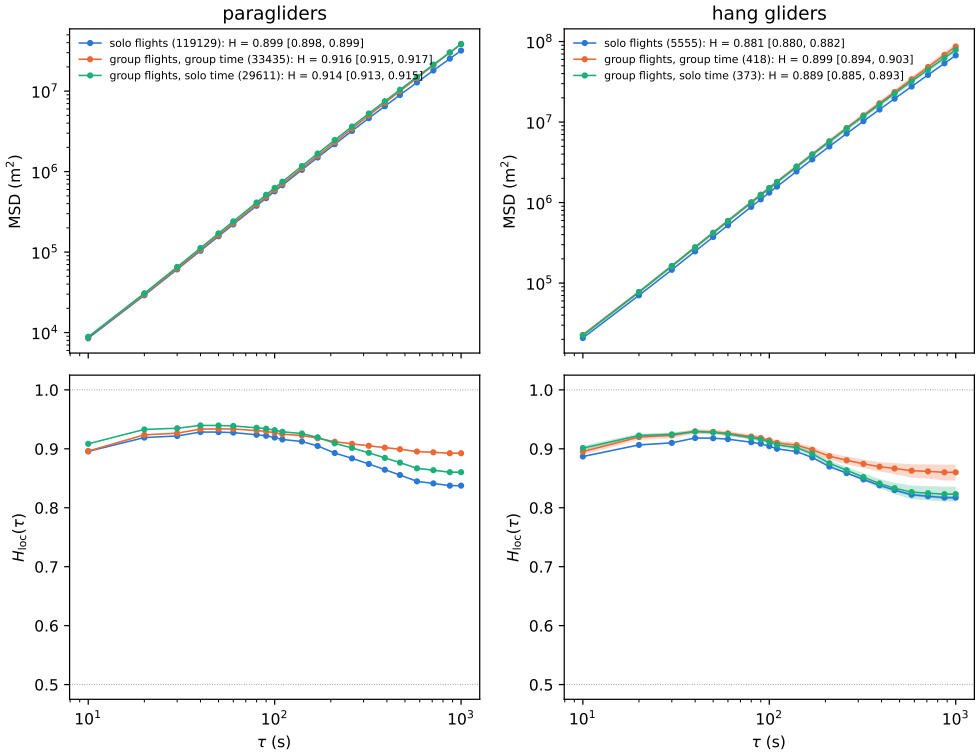


Figure 8: TA-MSD and H_{loc} of S, G and A, paragliders and hang gliders (hang gliders: 418 group flights, wide bands at long lags).

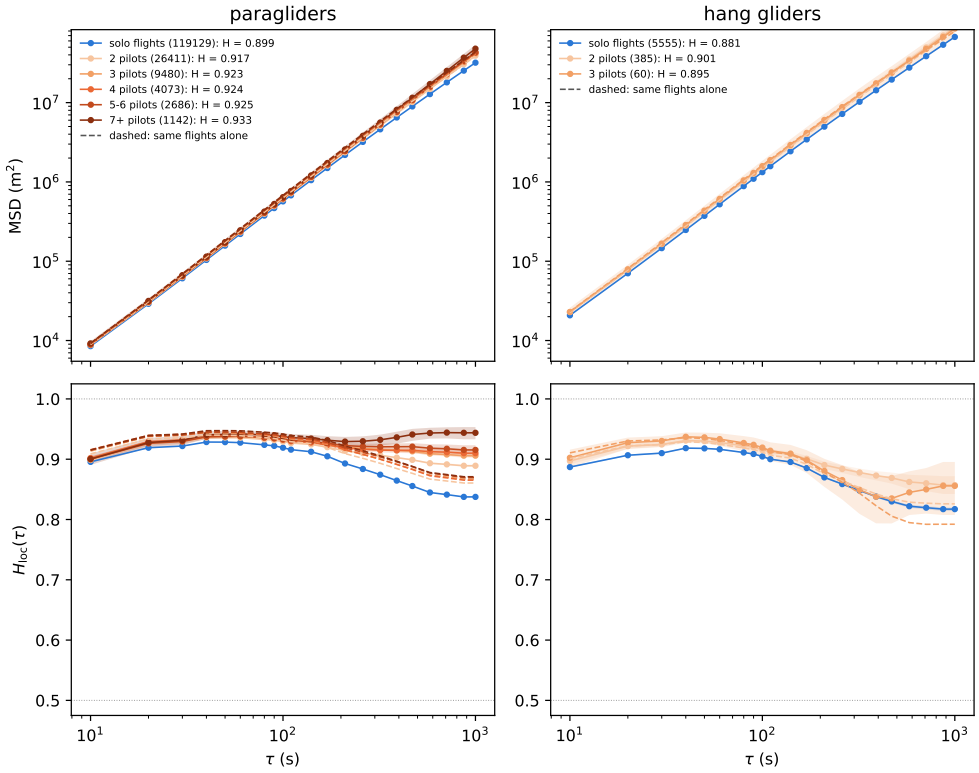


Figure 9: By the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

E Single site-days

Six site-days chosen by a fixed rule (most sizes with 15 flights or more, then most flights, at most two per takeoff site). Bands resample flights (too narrow: flights of one group are not independent).

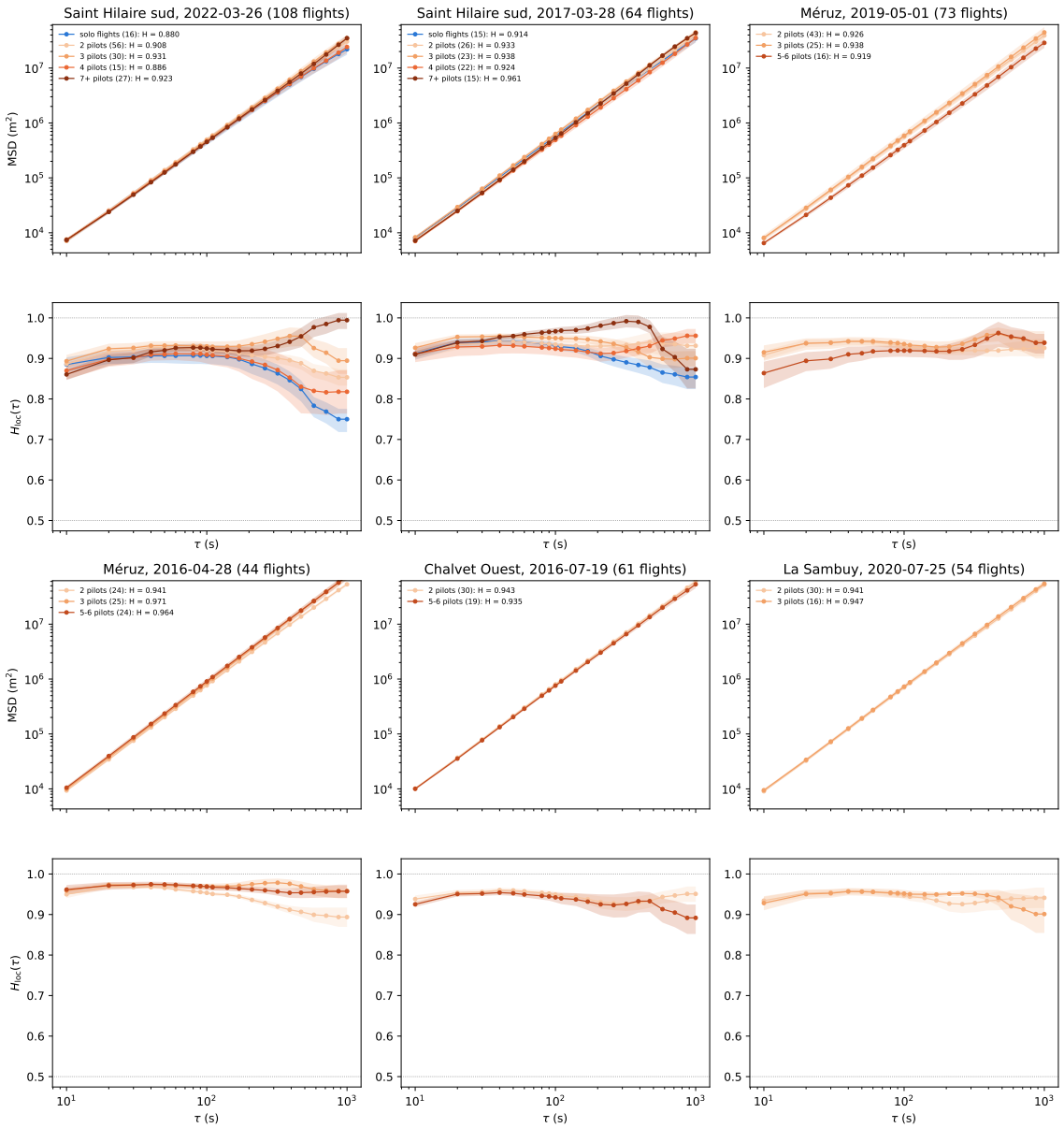


Figure 10: TA-MSD and H_{loc} on six site-days by the size of the configuration, with the solo flights of the same site-day where they reach 15 flights.

site-day	solo	2	3	4	5–6	7+
Saint Hilaire sud, 2022-03-26	0.750	0.853	0.894	0.818		0.994
Saint Hilaire sud, 2017-03-28	0.854	0.931	0.901	0.956		0.873
Méruz, 2019-05-01		0.926	0.939		0.938	
Méruz, 2016-04-28		0.894	0.958		0.957	
Chalvet Ouest, 2016-07-19		0.951			0.892	
La Sambuy, 2020-07-25		0.941	0.901			

Table 9: $H_{\text{loc}}(1000\text{ s})$ by size on each site-day (empty: fewer than 15 flights). Differences between days exceed those between sizes (the climb rate goes from 0.79 to 2.11 m/s across the six days).

F Robustness to the threshold

With $q = 0.2$ contacts need to last only 26.4 min: more and shorter groups, up to one contact in five chance.

	$q = 0.1$, $\tau^* = 38.1$ min	$q = 0.2$, $\tau^* = 26.4$ min
group flights (TA-MSD)	33 435	48 777
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, S	0.837	0.836
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, A	0.860	0.854
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, G	0.893	0.892
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, G − A same flight	+0.037 [+0.035, +0.039]	+0.032 [+0.029, +0.035]
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, G − A same s-d-h	+0.043 [+0.040, +0.046]	+0.042 [+0.040, +0.044]
$\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$, A − S same s-d-h	+0.003 [+0.001, +0.006]	+0.002 [−0.001, +0.005]
$\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$, G − A same flight	−0.009 [−0.009, −0.008]	−0.016 [−0.017, −0.015]
$\Delta H_{\mathrm{loc}}(100\ \mathrm{s})$, G − A same s-d-h	+0.002 [+0.001, +0.002]	−0.004 [−0.005, −0.003]
Δ searches to climb, G − A same s-d-h (points)	+2.6 [+2.3, +3.0]	+3.4 [+3.0, +3.7]
Δ climb rate, G − A same s-d-h (m/s)	+0.029 [+0.025, +0.034]	+0.024 [+0.020, +0.029]
Δ height lost per km, G − A same s-d-h (m/km)	−9.5 [−10.5, −8.6]	−11.5 [−12.2, −10.9]
Δ cross-country speed, G − A same s-d-h (km/h)	+1.29 [+1.16, +1.41]	+1.14 [+1.01, +1.30]

Table 10: The main results with the two thresholds (s-d-h: same site-day and hour). Every conclusion keeps its sign; at 100 s both estimates of the group effect are slightly negative with $q = 0.2$: no robust effect within one phase.

G Before, during and after with short windows

Same test as Section 6 with windows of 21 min, the shortest a piece of the TA-MSD allows. Groups of at least 21 min enter: more events, but at 1000 s a window holds about 26 origins, so H_{loc} over the longest lags is noisy and its drift between consecutive windows is large (controls below). The fitted H, dominated by short lags, shows no change beyond the drift; the phase statistics at entry depend on the window (time in climb +3.3 points, glide speed +0.63 km/h here, against +0.7 and +0.92 with 50 min).

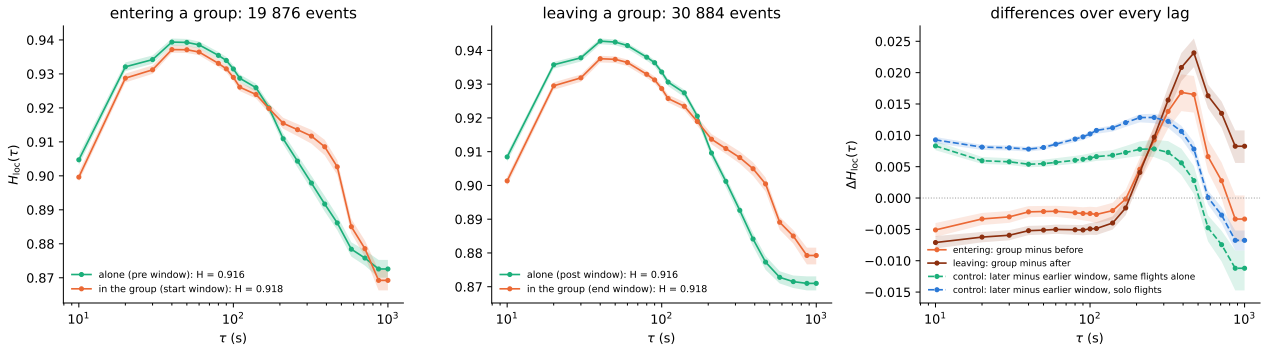


Figure 11: As Fig. 4, windows of 21 min.

comparison	events	ΔH	$\Delta H_{\mathrm{loc}}(100)$	$\Delta H_{\mathrm{loc}}(1000)$
entering: start − before	19 876	+0.002 [+0.001, +0.004]	−0.002 [−0.004, −0.001]	−0.003 [−0.007, +0.000]
leaving: end − after	30 884	+0.002 [+0.001, +0.003]	−0.005 [−0.006, −0.004]	+0.008 [+0.006, +0.011]
after − before	14 482	+0.004 [+0.003, +0.005]	+0.006 [+0.005, +0.008]	+0.002 [−0.001, +0.006]
control A: later − earlier	33 658	+0.006 [+0.005, +0.006]	+0.006 [+0.006, +0.007]	−0.011 [−0.015, −0.008]
control S: later − earlier	107 324	+0.009 [+0.009, +0.010]	+0.010 [+0.010, +0.011]	−0.007 [−0.008, −0.005]

Table 11: As Table 4, windows of 21 min.

comparison	events	time in climb (points)	searches to climb (points)	climb rate (m/s)	glide speed (km/h)	height lost per km (m/km)
entering: start − before	14 798	+3.3 [+2.9, +3.8]	+1.2 [+0.3, +2.2]	+0.043 [+0.035, +0.053]	+0.63 [+0.45, +0.77]	−8.9 [−12.1, −6.2]
leaving: end − after	23 014	+2.7 [+2.3, +3.1]	+0.9 [+0.2, +1.6]	+0.008 [+0.000, +0.018]	+0.54 [+0.42, +0.67]	−13.1 [−14.5, −11.5]
after − before	10 898	−1.5 [−2.0, −1.1]	−4.1 [−5.2, −2.9]	+0.017 [−0.001, +0.030]	+0.17 [−0.01, +0.35]	+9.6 [+5.7, +12.9]
control A: later − earlier	25 121	−2.1 [−2.4, −1.8]	−2.8 [−3.5, −2.0]	−0.001 [−0.009, +0.006]	−0.29 [−0.40, −0.15]	+3.4 [+1.9, +5.2]
control S: later − earlier	75 984	−2.0 [−2.2, −1.8]	−3.0 [−3.5, −2.7]	−0.012 [−0.017, −0.009]	−0.31 [−0.37, −0.24]	+6.2 [+5.0, +7.5]

Table 12: As Table 5, windows of 21 min.

H Leader minus follower

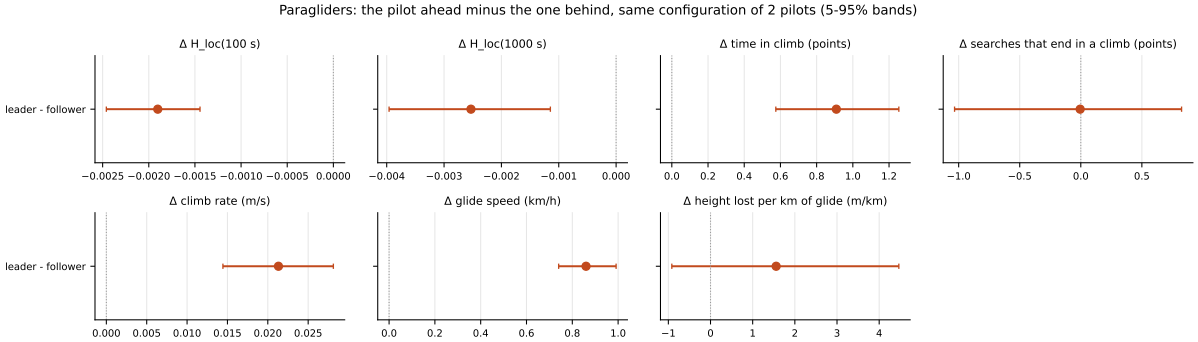


Figure 12: Leader minus follower, same configuration of two pilots.

Δ	leader − follower
$H_{\text{loc}}(100\ \text{s})$	−0.002 [−0.002, −0.001]
$H_{\text{loc}}(1000\ \text{s})$	−0.003 [−0.004, −0.001]
time in climb (%)	+0.91 [+0.57, +1.25]
searches ending in a climb (%)	−0.01 [−1.03, +0.83]
climb rate (m/s)	+0.021 [+0.014, +0.028]
glide speed (km/h)	+0.86 [+0.74, +0.99]
height lost per km of glide (m/km)	+1.6 [−0.9, +4.5]
cross-country speed (km/h)	+0.14 [+0.11, +0.16]

Table 13: Leader minus follower, same configuration of two pilots (difference with 5–95% band).

I Reproducing the report

From `xc-lab/analysis/groups/group-identification`, SSD_DISANTE mounted:

```
uv run python tamsd.py; uv run python phase_msd.py; uv run python survival.py
uv run python budget.py; uv run python performance.py; uv run python blocks.py
uv run python redundancy.py; uv run python size.py; uv run python decompose.py
uv run python zones.py; uv run python prepost.py; uv run python prepost.py --window 1260
uv run python mechanisms.py; uv run python report/facts.py
uv run python report/tables.py; cd report; latexmk -pdf group_vs_solo.tex
```

Heavy pieces stay on SSD_DISANTE, in the `analyses/` folder of each run; figures and JSON in `out/<run>/`.